

An Internship Report
Of
IN-MEMORY CACHE WITH TTL LITE
Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B-Tech (CSE)



By
Roll Number of Student:- 2410030747
Name of Student: Manish Kumar
Batch: 2026-2

CANDIDATE'S DECLARATION

I, Manish Kumar, do hereby declare that the internship report titled "In-Memory Cache with TTL Lite" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

All references have been properly acknowledged and cited. This report is my original work and has not been submitted to any other institution for any degree or diploma.

I further declare that I shall be solely responsible for any copyright violation, if any, arising from this work.

Signature :

Student Name: Manish Kumar

Roll No: 2410030747

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to AICTE and EduSkills Academy for providing me with the opportunity to undertake the Virtual Internship Program in the domain of MERN Full Stack Development.

I am highly thankful to my industry mentor, **Ms. Sagarabala Behera**, for her valuable guidance, continuous support, and technical assistance throughout the internship period. Her insights and encouragement helped me successfully complete my project titled "**In-Memory Cache with TTL Lite.**"

I also express my heartfelt thanks to the faculty members of the School of Computer Science and Engineering, IILM University, Greater Noida, for their support and encouragement.

Finally, I thank my parents and friends for their motivation and constant support during the internship.

Signature :

Student Name: Manish Kumar

Roll No: 2410030747

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3. INTERNSHIP COMPLETION CERTIFICATE

CERTIFICATE
OF INTERNSHIP







PRESENTED TO

Manish Kumar

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

MERN full Stack Development With Project

During the internship, the learner demonstrated proficiency in:

- Welcome to Modern Web Development & Project Setup & Core JavaScript & Web Concepts for Full-Stack
- React Fundamentals - Building User Interfaces & React Hooks and Side Effects
- State Management in React (Context API & Redux Toolkit) & React Router for Navigation
- Styling React Applications (TailwindCSS) & Project 1 - Building a React Single Page Application
- Node.js and Express.js Fundamentals & Database Integration with MongoDB and Mongoose
- Building RESTful APIs with Express.js & Authentication and Authorization (JWT)
- Full-Stack Integration - Connecting React and Node.js & DevOps Basics - Git Workflow and Deployment & Advanced React Patterns and Performance & Advanced Node.js & Backend Best Practices
- Capstone Project: E-commerce Platform with User Management & Product Catalog
- **Project: In-Memory Cache with TTL Lite**

This certification recognizes the successful completion of the internship requirements and project work.



Issued on: Aug 25, 2026
Certificate ID: 2026-757F405765



Mitu Swain
General Manager
(Learning & Development)
EduSkills

Internship Offer Letter



Ref No: C17Y2026M081525488

Internship Offer Letter

Student Name : Manish Kumar
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : MERN full Stack Development + Project
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU69e664fd032951776706813

We are pleased to inform you that you have been selected for the 8-Week AICTE – EduSkills Virtual Internship Program.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Welcome to Modern Web Development & Project Setup & Core JavaScript & Web Concepts for Full-Stack	Course introduction, environment setup, project initialization, and strengthening foundational JavaScript and essential web development concepts.
Week 2	React Fundamentals – Building User Interfaces & React Hooks and Side Effects	Learn JSX, components, props, state, and advanced hooks like useEffect and custom hooks for reusable logic.
Week 3	State Management (Context API & Redux Toolkit) & React Router for Navigation	Implement global state management and build client-side routing for seamless navigation in React applications.
Week 4	Styling React Applications (TailwindCSS) & Building a React Single Page Application	Style applications using Tailwind CSS and build a complete SPA applying routing and state management.
Week 5	Node.js & Express.js Fundamentals & Database Integration with MongoDB & Mongoose	Introduction to backend development, server creation, and integrating MongoDB using Mongoose for efficient data management.
Week 6	Building RESTful APIs with Express.js & Authentication and Authorization (JWT)	Design secure REST APIs and implement authentication and authorization using JSON Web Tokens.
Week 7	Full-Stack Integration & DevOps Basics – Git Workflow & Deployment & Advanced React Patterns & Advanced Node.js Best Practices	Connect frontend and backend, implement Git workflows and deployment, and explore advanced scalable frontend and backend techniques.
Week 8	Capstone Project – E-commerce Platform with User Management & Product CatalogProject	Build a complete full-stack e-commerce platform with authentication, product management, API integration, shopping cart, deployment, and final project implementation.
Project/Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



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4. PROJECT DESCRIPTION

4.1 Introduction

Internships provide practical exposure to real-world technologies and help bridge the gap between academic learning and industry requirements. Through industry-oriented training, students gain experience in applying theoretical concepts to practical software development problems.

As part of the AICTE–EduSkills Virtual Internship Program, I worked on the project titled **"In-Memory Cache with TTL Lite."**

The project focuses on designing and implementing a lightweight caching system capable of storing and retrieving data efficiently while automatically removing expired entries through a Time-To-Live (TTL) mechanism.

This project demonstrates important software engineering concepts such as memory management, data structures, cache optimization, and performance monitoring.

4.2 Organization Profile

Organization Name

EduSkills Academy

Domain

EdTech and Skill Development

About the Organization

EduSkills Academy is a leading skill-development platform that collaborates with educational institutions and industry partners to provide students with industry-relevant training and internship opportunities.

The organization focuses on enhancing employability by offering practical learning experiences in emerging technologies and software development domains.

4.3 Problem Statement

Modern software applications require quick access to frequently used data. Repeated retrieval of information from databases increases response time and puts additional load on system resources.

The project "In-Memory Cache with TTL Lite" addresses this challenge by implementing a lightweight cache that stores frequently accessed data in memory for faster retrieval.

The system includes:

- SET operation
- GET operation
- TTL-based expiration
- DELETE functionality
- Cache statistics generation
- Performance monitoring through hit and miss tracking

This ensures efficient data management and improved application performance.

4.4 Project Objectives

The main objectives of the project are:

1. To design and implement an efficient in-memory cache.
2. To provide support for SET and GET operations.
3. To implement Time-To-Live (TTL) functionality.
4. To automatically remove expired data.
5. To allow manual deletion of cache entries.
6. To track cache hits and misses.
7. To generate cache statistics reports.
8. To understand practical applications of caching systems.

4.5 Scope of the Project

The scope of the project includes:

- Fast storage and retrieval of data.
- Temporary storage of frequently used information.

- TTL-based automatic expiration.
- Manual cache management.
- Performance tracking and reporting.
- Understanding real-world cache implementations.

The project serves as a foundational implementation of caching systems used in web applications, distributed systems, and cloud computing environments.

4.6 Technologies and Tools Used

Programming Language

- JavaScript

Development Tools

- Node.js
- Visual Studio Code
- Git
- GitHub

Concepts Used

- In-Memory Data Structures
- Time-To-Live (TTL)
- Cache Management
- Data Retrieval
- Performance Monitoring

Technologies Learned During Internship

- HTML
- CSS
- JavaScript
- React.js
- Redux Toolkit
- React Router
- Tailwind CSS
- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT Authentication

4.7 System Architecture

Brief Description

The In-Memory Cache with TTL Lite system is designed to store and retrieve data efficiently using an in-memory caching mechanism. The architecture consists of multiple modules that work together to manage cache operations, automatically remove expired data, track cache performance, and generate statistics. This structure ensures fast data access while maintaining data validity through Time-To-Live (TTL) functionality.

System Components

- SET Module
 - Stores key-value pairs in cache memory.
 - Adds new entries and updates existing entries.
- GET Module
 - Retrieves data using a unique key.
 - Returns cached data if available and valid.
- TTL (Time-To-Live) Manager
 - Assigns expiration time to cache entries.
 - Automatically invalidates expired data.
- DELETE Module
 - Removes specific entries from the cache.
 - Helps in manual cache management.
- Statistics Module
 - Tracks cache hits and cache misses.
 - Monitors overall cache performance.
- Report Generation Module
 - Generates cache statistics reports.
 - Displays cache size, hit count, and miss count.

4.8 Methodology

Phase 1 – Basic Cache Implementation

Implemented SET and GET operations for storing and retrieving data.

Phase 2 – TTL Integration

Added Time-To-Live functionality to automatically expire cache entries.

Phase 3 – DELETE Operation

Implemented cache deletion functionality for manual removal of data.

Phase 4 – Performance Monitoring

Introduced hit and miss counters to measure cache efficiency.

Phase 5 – Statistics Reporting

Developed a reporting mechanism that displays cache size, hits, and misses.

Phase 6 – Testing

Verified all operations and validated cache performance.

4.9 Expected Outcomes

The expected outcomes of the project are:

- Efficient storage and retrieval of data.
- Improved application performance.
- Automatic removal of expired entries.
- Better understanding of cache management.
- Performance monitoring through statistics.
- Development of programming skills.
- Exposure to industry-standard software development practices.

The project successfully demonstrates the practical implementation of caching concepts.

4.10 Certificates of Completion and Communication Proof

The following documents are attached:

1. AICTE–EduSkills Internship Completion Certificate.
2. Offer Letter
3. Communication Emails

These documents verify the successful completion of the internship and project work.

AICTE-EduSkills Internship Certificate download link External Inbox x

EduSkills Foundation
to me ▾

Tue, Aug 25, 12:32 PM (3 days ago)

☆ 😊 ↶



Up next

Dear **Manish Kumar**,

🎉 **Congratulations!**

You have successfully completed **10 weeks MERN full Stack Development + Project**. Your certificate is ready — please log in to your profile and download it.

Steps to download your certificate:

1. Login at: <https://eduskillsfoundation.org/login>
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Please post your certificate on your LinkedIn profile to get better job opportunities and visibility from corporates.

5. BIBLIOGRAPHY/REFERENCES

Mozilla Developer Network (MDN) JavaScript Documentation.

Node.js Official Documentation.

Express.js Official Documentation.

MongoDB Official Documentation.

AICTE–EduSkills Virtual Internship Learning Resources.

Technical articles related to:

- In-Memory Caching
- Time-To-Live (TTL)
- Cache Performance Monitoring
- Software Development Best Practices